

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B.Tech. Branch : Mechanical Engineering Semester : V Subject Code : BTAPE504D Subject Name: Automobile Engineering Max Marks: 60 Date: 17/08/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		
A)	What are the basic components of an automobile system explain any four components in details?	1	6M
B)	What are the basic cylinder arrangements for automotive engines? Explain any three arrangements in detail?	1	6M
C)	What factors affect the tractive force requirement of a vehicle? Also explain vehicle performance curve?	1	6M
Q.2	Solve Any Two of the following.		
A)	Discuss in brief front axle? Explain any one type with neat sketch?	2	6M
B)	Define the following:- (1) Caster & Camber ? (2) toe-in & toe-out (3) king pin inclination.	2	6M
C)	Discuss in brief with neat sketch steering system layout.	2	6M
Q.3	Solve Any Two of the following.		
A)	What are the benefits of vehicle safety system? How many types of safety are there in car?	3	6M
B)	What are the different types of safety glass? Also mention properties and applications of safety glass?	3	6M
C)	Discuss in brief different types of car mirrors?	3	6M
Q.4	Solve Any Two of the following.		
A)	What are the basic requirement of wheels and tyres? Classify road wheels? Explain any one type with neat sketch.	4	6M
B)	Explain in brief wheel balancing? Also explain cross section of tubless tyre?	4	6M
C)	What are various factors affecting Tyre life and performance?	4	6M
Q.5	Solve Any One of the following.		
A)	Explain in brief Battery charging system? Also explain maintenance of lead acid batteries?	5	6M
B)	Explain in detail need of vehicle testing? Also explain vehicle test standards?	5	6M
C)	Discuss in brief emission control system and regulations.	5	6M
*** BEST OF LUCK ***			

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